

A Diagnostic Check on a One-Parameter Matter-Exponent Deformation of Λ CDM

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DESI DR2 baryon acoustic oscillation (BAO) data, combined with Type Ia supernovae, have been read within two-parameter equation-of-state templates as a preference for dynamical dark energy, with a strength that depends sharply on the supernova catalog. We test a minimal one-parameter deformation of the flat Λ CDM expansion history, $H^2/H_0^2 = \Omega_m(1+z)^\gamma + \Omega_\Lambda$ (with $\gamma = 3$ recovering Λ CDM), not as a candidate cosmology but as a diagnostic of how sensitive such a preference is to the very-low-redshift supernova sample. Using Pantheon+ (1701 SNe Ia with full covariance) and DESI DR2 BAO (13 points), with the design and priors frozen before fitting, the full supernova vector gives $\gamma < 3$ at $\Delta\chi^2 \approx -8.75$ relative to Λ CDM. The standard $z > 0.01$ cosmology cut, adopted in the supernova-systematics literature for peculiar-velocity and calibration reasons, reduces this to $\Delta\chi^2 \approx -2.97$, and a $z > 0.1$ cut to -1.89 . A Savage-Dickey comparison favors Λ CDM under both a narrow ($\gamma \in [2, 4]$) and a wide ($\gamma \in [0, 5]$) prior ($\Delta \ln Z = -1.09$ and -2.00), and the same cut heals a previously disjoint supernova-only versus BAO-only split. The large apparent preference is therefore low-redshift driven, leaving a mild residual ($\hat{\gamma} \approx 2.90$, $\Delta\chi^2 \approx 3.0$) consistent in scale with published mild Pantheon+ preferences. We offer it as a small independent data point for the discussion of low-redshift supernova systematics.

I. MOTIVATION

The DESI DR2 BAO measurements [1], combined with Type Ia supernovae and cosmic microwave background priors, have been interpreted within two-parameter equation-of-state parameterizations as a preference for dynamical dark energy over a cosmological constant. A now well-documented feature of that preference is its dependence on the supernova catalog: it is strongest for the DES five-year sample, weaker for Union3 [2], and weakest, below 2σ , for Pantheon+ [3]. Efstathiou [4] traced much of the spread to an offset of order 0.04 mag between the low- and high-redshift distance scales of supernovae common to the DES five-year and Pantheon+ compilations; Huang, Cai, and Wang [5] tied the DESI-era evidence substantially to the low-redshift sample, which Vincenzi *et al.* [6] reanalyzed in detail. Chaudhary *et al.* [7], in an independent late-time-only analysis, find a dataset-dependent preference that stays below 2σ for Pantheon+ but rises to 2 – 2.74σ with Union3 or DES-Dovekie [8], consistent with a Quintom-B direction ($w_0 > -1$, $w_a < 0$).

Against that backdrop we examine a deliberately minimal object: a one-parameter deformation of the flat Λ CDM background,

$$\frac{H^2(z)}{H_0^2} = \Omega_m(1+z)^\gamma + \Omega_\Lambda, \quad \gamma = 3 \text{ recovers } \Lambda\text{CDM}. \quad (1)$$

Softening the matter exponent below $\gamma = 3$ reproduces, at the level of the late-time expansion shape, the same reduced high-redshift growth of H^2 that the two-parameter

fits capture as $w_a < 0$. We do not advance Eq. (1) as a physical cosmology. We use it as a single-parameter diagnostic: a sharp, low-dimensional instrument for asking how much of an apparent late-time departure from Λ CDM lives in the very-low-redshift supernova sample that the systematics literature has flagged.

II. MODEL, DATA, AND FROZEN DESIGN

The model roster, priors, data selection, and pass/fail thresholds were fixed in a version-controlled commit before any fitting, and a single results commit followed; the git history is the audit trail. Constraints use Pantheon+ (1701 SNe Ia with the full distance covariance [3]) and DESI DR2 BAO (13 points [1]), with the same integrator, redshift grid, and nuisance treatment (M_B and $H_0 r_d$ handled identically across all models) as the pipeline of Ref. [9], which reproduces the flat- Λ CDM baseline $\chi^2 = 1772.45$ on these data. Redshift cuts act on the supernova data vector and its covariance through a single index mask, with rows and columns subselected together and no re-derivation of the covariance.

The exponent prior is flat on $[2, 4]$ (primary; symmetric about Λ CDM), with a wider flat $[0, 5]$ prior reported for robustness; Ω_m is flat on $[0, 1]$. For context we fit, in the same pipeline, a standard two-parameter roster (the $w_0 w_a$ CDM/CPL, BA, and JBP parameterizations, with $w_0 \in [-3, 1]$ and $w_a \in [-3, 2]$) and, as a separate diagnostic, a curved $o\Lambda$ CDM model. Evidence for the nested hypothesis $\gamma = 3$ is computed from the exponent posterior by the Savage-Dickey density ratio, with a bandwidth-scan stability check, under both frozen priors; following the convention we compare against [7], we read $|\Delta \ln Z| < 1$ as inconclusive and 1 – 3 as moderate.

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TABLE I. The improvement over Λ CDM collapses as the low-redshift supernovae are removed; the pre-registered bar after the $z > 0.01$ cut was $\Delta\chi^2 \leq -4$.

Sample	N_{SN}	$\Delta\chi^2 (\gamma\text{-CDM} - \Lambda\text{CDM})$	$\hat{\gamma}$
full	1701	-8.75	2.82
$z > 0.01$	1590	-2.97	2.90
$z > 0.1$	960	-1.89	2.90

III. RESULTS

Three counts organize the result: the shipped Pantheon+ vector carries $N_{\text{all}} = 1701$ light curves, of which $N(z > 0.01) = 1590$ and $N(z > 0.1) = 960$ survive the two cuts. The 111 supernovae below $z = 0.01$ are exactly those the published Pantheon+ cosmology analysis excludes on peculiar-velocity and calibration grounds.

On the full vector the deformation prefers $\gamma < 3$ at $\Delta\chi^2 = -8.75$ relative to Λ CDM, for one extra parameter. The standard $z > 0.01$ cut reduces this to $\Delta\chi^2 = -2.97$; the $z > 0.1$ stress cut reduces it further to -1.89 (Table I). After the $z > 0.01$ cut the exponent posterior is $\hat{\gamma} = 2.90$ with a 68% interval [2.836, 2.956]: the interval excludes $\gamma = 3$, but the χ^2 improvement no longer clears the pre-registered $\Delta\chi^2 \leq -4$ bar for a low-redshift-vetted detection.

Bayesian model comparison points the same way. The Akaike criterion gives $\Delta\text{AIC} = -0.97$ post-cut, short of the -2 bar, and the best member of the two-parameter roster (w CDM) reaches only $\Delta\text{AIC} \approx -2.1$, so nothing in the roster beats Λ CDM after the cut. The Savage-Dickey ratio favors Λ CDM under both priors, $\Delta \ln Z = -1.09$ (primary [2, 4]) and -2.00 (robustness [0, 5]); the wider prior gives the more conservative reading, since a narrow prior mechanically flattens the deformation's evidence, and yields a moderate preference for Λ CDM. That the posterior excludes $\gamma = 3$ at 68% while the evidence favors the nested model is the expected prior-volume behavior.

Two further diagnostics complete the picture. First, the supernova-only and BAO-only exponent preferences, disjoint on the full vector, overlap once the low-redshift subset is removed: the 68% intervals become [2.31, 3.15] (supernovae) and [2.85, 3.02] (BAO), so the earlier split was itself sensitive to the very-low-redshift subset rather than a broad supernova-versus-BAO conflict. Second, on the full vector the CPL template image of the de-

formation lies in the community's direction (a softened late-time shape, $w_a < 0$); after the $z > 0.01$ cut it flips to $(w_0, w_a) = (-0.97, +0.36)$, with $w_a > 0$, opposite to the direction of the two-parameter DESI fits. A curvature check finds Λ CDM capturing only 47% of the deformation's full-vector improvement, below the 70% flag: curvature is not a substitute for the same shape.

IV. DISCUSSION AND CONCLUSION

The reconciliation is the main result. On the full Pantheon+ vector the deformation looked economical, $\Delta\chi^2 \approx 8.75$ for a single parameter, well above the mild preferences ($\Delta\chi^2 \approx 1.7$ – 1.9 , below 2σ) that published Pantheon+ analyses report for dynamical dark energy. Almost all of that gap closes in a single cut, and with it the supernova-versus-BAO split and the template alignment. Stated plainly: The same cut that brings the analysis into alignment with standard Pantheon+ cosmology practice also removes most of the apparent soft-exponent signal.

We keep the language conservative: the apparent preference is low-redshift driven, or equivalently sensitive to the very-low-redshift subset, and the reader can connect that sensitivity to the peculiar-velocity and calibration considerations behind the standard $z > 0.01$ cut, examined for the low-redshift sample by Efstathiou [4], Huang *et al.* [5], and Vincenzi *et al.* [6]. Lee [10] separately cautions that CPL-type templates can mislead at high redshift, consistent with the template sign behavior we recover.

A mild residual survives every cut: $\hat{\gamma} \approx 2.90$ at $\Delta\chi^2 \approx 3.0$ after the $z > 0.01$ cut and ≈ 1.9 after $z > 0.1$, with the evidence still favoring Λ CDM. It is consistent in scale with mild Pantheon+ preferences already in the literature, and worth one sentence rather than a paper. The one-parameter deformation of Eq. (1) is therefore not advanced as a cosmology. Its value is as a diagnostic: a soft-matter-exponent preference here is substantially a property of the very-low-redshift supernova sample rather than of the late-time expansion history. We offer it as a small independent contribution to the ongoing discussion of low-redshift supernova systematics. *Data availability.* The fitting pipeline is archived in Ref. [9]; the pre-registered design and the outputs quoted here are recorded in the same analysis repository at commits 343c9d8 (design) and 69c4604 (results/gamma_gate_*).

[1] DESI Collaboration, arXiv:2503.14738 (2025).
[2] D. Rubin *et al.*, arXiv:2311.12098 (2023).
[3] D. Brout *et al.*, *Astrophys. J.* **938**, 110 (2022).
[4] G. Efstathiou, *Mon. Not. R. Astron. Soc.* **538**, 875 (2025).
[5] L. Huang, R.-G. Cai, and S.-J. Wang, arXiv:2502.04212 (2025).

[6] M. Vincenzi *et al.* (DES Collaboration), arXiv:2501.06664 (2025).
[7] H. Chaudhary, V. K. Sharma, S. Capozziello, and G. Mustafa, arXiv:2510.08339 (2026).
[8] B. Popovic *et al.*, arXiv:2506.05471 (2025).
[9] B. Shatto, Acos, doi:10.5281/zenodo.19798852 (2026).
[10] S. Lee, arXiv:2506.18230 (2025).